RUPAYAN SARKAR

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Github | Portfolio | Kolkata, West Bengal, India

SUMMARY

ECE undergraduate skilled in Embedded Systems, Robotics, IoT, and Python, with hands-on experience in Arduino, NodeMCU, PID control, CNC systems, and sensors. Strong foundation in DSA, OOP, Computer Networks, and Digital Electronics.

EDUCATION

Cooch Behar Government Engineering College — Cooch Behar, West Bengal
B.Tech in Electronics & Communication Engineering | CGPA: 7.00 (up to 3rd Year) — *Sep 2023 – Jun 2027*

Rahara Ramakrishna Mission Boys' Home — Khardaha, West Bengal
Higher Secondary (WBCHSE) | Score: 70% — *2023*

Rahara Ramakrishna Mission Boys' Home — Khardaha, West Bengal
Secondary (WBBSE) | Score: 89% — *2021*

INTERNSHIP EXPERIENCE

Summer Intern – Microelectronics Technology & VLSI Design 2026 (July–August) at **IC Centre, Department of Electronics & Telecommunication Engineering, Jadavpur University.**
Currently pursuing the Summer Internship Programme on Microelectronics Technology and VLSI Design at Jadavpur University, gaining practical exposure to VLSI design, microelectronics, and semiconductor technologies through laboratory and research-oriented work.

PROJECTS

CNC Plotter (Handwriting Machine) [GitHub]
Arduino Uno R3, GRBL Firmware, G-Code
- Built a CNC plotter with ±0.5 mm drawing accuracy using Arduino Uno and GRBL.
- Implemented G-Code interpretation and calibrated stepper motors for precise motion.

Line Following Robot
PID Control, Embedded Systems, Autonomous Navigation
- Designed a PID-controlled autonomous robot with optimised path tracking.
- Demonstrated reliable navigation at College Tech-Fest 2025.

Droid Blitz Robot
NodeMCU, Mecanum Wheels, IoT, Wireless Control
- Developed a Mecanum-wheel robot with smartphone-based wireless control.
- Competed in robotics events and conducted a hands-on student workshop.

Earthquake Indicator
Sensor Integration, Data Processing, Embedded Systems
- Built an accelerometer-based earthquake monitoring system.
- Advanced to the Smart India Hackathon 2024 screening round.

Robotic Arm
Robotics, Additive Manufacturing, Mechanical Design
- Contributed to the design and assembly of a robotic arm for additive manufacturing applications.

RESPONSIBILITIES

Communication Head – Roboverse Club, CGEC
- Led coordination of technical/outreach events and increased student engagement in robotics.
- Co-organised a two-day seminar on Designing Intelligent Systems (DSTBT, Govt. of West Bengal).
- Arranged the Techno Commercial Event at Byte-Burst (2025–26) and conducted a Droid Blitz workshop.

TECHNICAL SKILLS

Languages: C, Python, HTML, CSS
Embedded Systems & Robotics: Arduino UNO, NodeMCU, ESP8266, PID Control, IoT, CNC Systems, Sensors & Actuators
Tools: Git, GitHub, VS Code, GitHub Actions
Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Digital Electronics, Microprocessors

CERTIFICATIONS

- Arduino and Embedded Systems Training
- 3D Printing Internship & Training (2026)
- Smart India Hackathon 2024 – Participation Certificate

LANGUAGES

Bengali (Native) | English (Professional) | Hindi (Professional)